

Assignment 3 - Developing a publishable research proposal

Submission Date: Sunday, October 4th, 2026, 11:59pm

Class presentations: Monday, October 5th, 2026

Apply the multi-agent research workflow developed in 05_01 and 05_02 to a GeoAI research problem of your own interest that you would like to write a paper on. Use the research question, critic, writer, methodology, data/tool and validation roles to develop and refine your proposal, retaining the intermediate outputs and your own review decisions.

Importantly, these tools are there to support you, but are not substitutes for your own critical thinking about this topic (e.g., you are the human-in-the-loop needing to direct and make executive strategic decisions).

Create a 5-minute presentation proposing your topic to the class. Present three possible research questions refined with AI, but it is essential you manually review them, and critique each for scientific value and feasibility yourself. Select the strongest question and explain your choice, including which AI suggestions you accepted, changed or rejected and why.

Develop a workable method for the selected question, scoped to a paper that could be completed and submitted to a peer-reviewed journal within the next eight weeks using AI-assisted coding and analysis. The emphasis is on a credible research concept and feasible plan; completed research results are not required for this assignment.

Table 1 Proposal aspects to address.

Component	Question to address
Topic and questions	What problem matters, and what are your three refined, testable research questions? Explain your own feasibility critique and preferred question.
Literature and contribution	Include at least five supporting peer-reviewed references from the last 2-3 years (e.g., very recent papers/the state-of-the-art), to serve as a foundation to build on. Explain what your proposed work would enhance so you have a novel contribution. Read and verify the sources yourself.
Method and feasibility	Identify data and access, study area/unit of analysis, analytical steps, AI-assisted coding tools, a baseline/comparator and validation checks. Address spatial assumptions, key risks and what evidence could weaken your claim. Give an eight week work plan.
Journal target	Name a suitable peer-reviewed journal with a Journal Impact Factor greater than 3 (basically, a reputable journal). Report the metric year and source, and explain why your proposed paper fits its scope.
Subject matter expert	Identify at least one potential co-author within the department or elsewhere. Give their affiliation, relevant expertise and the scientific content checks they could provide. You are identifying a possible collaborator, not claiming their agreement.

Please submit your presentation online as a PDF by 11:59pm on Sunday, October 4th. You will present it at the next class on Monday, October 5th. Use supporting slides for full references, the detailed work plan and examples of agent outputs and your manual revisions.

Prepare professional, legible slides with consistent formatting, slide numbers, source acknowledgments and an AI disclaimer identifying the tools used, their contributions and your own checks. Proofread for spelling and grammatical errors. Rehearse a clear, suitably paced five minute explanation. Your presentation approach, style and delivery will also be graded.

Past presentations this semester have occasionally been below acceptable presentation standards, and will be penalized in future assignments (covering both slide quality and presenter style/oral talking points). Assume this is a job interview and your professional future depends on it.

Table 2 Assessment criteria.

Criterion	Evidence	Weight
Research concept and literature	Significant topic, three clear questions, a justified contribution grounded in at least five references, and a suitable journal target.	30%
Method and feasibility	Workable data, methods, validation and eight week plan; realistic use of AI-assisted agentic coding/reasoning and an appropriate potential expert co-author.	30%
Critical human review	Evidence of the multi-agent workflow, manual critique of all three questions, and reasoned decisions to accept, revise or reject AI suggestions.	20%
Presentation quality and delivery	Professional slides, slide numbers, citations, AI disclaimer and careful proofreading; clear structure, effective delivery and adherence to five minutes.	20%

Students remain responsible for the accuracy and defensibility of all submitted material. Verify citations and distinguish proposed contributions from established findings. Agreement between agents is not independent scientific validation.